INCH-POUND

MIL-M-28787/345
AMENDMENT 1
30 September 1992

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CONVERTER, S/D, KEY CODE JAP

This amendment forms a part of Military Specification MIL-M-28787/345(EC), dated 14 November 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

3.5.3: Delete "45 grams" and substitute "50 grams".

PAGE 4

TABLE I, Test condition G1, Min column: Delete "1.82" (5 places) and substitute "1.05" (5 places).

TABLE I, Test condition G1, Max column: Delete "3.38" (5 places) and substitute "1.95" (5 places).

TABLE I, Test condition G2, Min column: Delete "1.96" (5 places) and substitute "0.43" (5 places).

TABLE I, Test condition G2, Max column: Delete "3.64" (5 places) and substitute "0.80" (5 places).

PAGE 5

TABLE I: Delete and substitute new table I as printed on page 3 of this amendment.

PAGE 6

TABLE I: Delete and substitute new table I as printed on page 4 of this amendment.

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PAGE 9

FIGURE 2: Delete and substitute new figure 2 as printed on page 6 of this amendment.

PAGE 18

TABLE VII, Test condition G1, Min column: Delete "1.82" and substitute "1.05".

TABLE VII, Test condition G1, Max column: Delete "3.38" and substitute "1.95".

TABLE VII, Test condition G2, Min column: Delete "1.96" and substitute "0.43".

TABLE VII, Test condition G2, Max column: Delete "3.64" and substitute "0.80".

PAGE 43

Delete and substitute new page 43 as printed on page 7 of this amendment.

PAGE 45

FIGURE 19, NOTE 2: Delete "PW = 2 μs $\pm$ 10%." and substitute "PW = 2 sec $\pm$ 10%."

PAGE 52

TABLE XXIII: Delete and substitute new table XXIII as printed on page 8 of this amendment.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Preparing activity:
Navy - NW

(Project 5963-0260)

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
L	Sensitivity check	XI	S	0.010	0.033	0.010	0.033	0.010	0.033	0.010	0.033	0.010	0.033	Degree
M1-M14	Digital output voltage (Voh)	XII	X	2.4	5.1	2.4	5.1	2.4	5.1	2.4	5.1	2.4	5.1	V
N1-N14	Digital output voltage (Vol)	XIII	X	0	400	0	400	0	400	0	400	0	400	mV
P1-P3	Digital input breakdown current	XIV	S	0	40	0	40	0	40	0	40	0	40	μA
Q1-Q3	Digital input current	XV	S	-1.6	0.0	-1.6	0.0	-1.6	0.0	-1.6	0.0	-1.6	0.0	mA
R1-R16	Three-state leakage current, high level	XVI	X	-1	20	-1	20	-1	20	-1	20	-1	20	μA
S1-S16	Three-state leakage current, low level	XVII	X	1	-20	1	-20	1	-20	1	-20	1	-20	μA
T	Delay time	XVIII	Q	0	3	0	3	0	3	0	3	0	3	μs
U1, U2	Delay time	XVIII	Q	0	150	0	150	0	150	0	150	0	150	ns
V1, V2	Tracking rate check	XIX	Q	10	15	10	15	10	15	10	15	10	15	rps

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
W	Settling time	XX	Q	0	200	0	200	0	200	0	200	0	200	ms
X	Noise sensitivity	XXI	Q	← See table XXI →										
Y1	Converter busy, direct (VH)	XXII	X	2 4	5.1	2.4	5.1	2.4	5.1	2.4	5.1	2.4	5.1	V
Y2	Converter busy, direct (VL)	XXII	X	0	400	0	400	0	400	0	400	0	400	mV
Y3	Converter busy, direct (PW)	XXII	X	0 5	6.0	0.5	6.0	0.5	6.0	0.5	6.0	0.5	6.0	µS
Y4	Converter busy, 3-state (VH)	XXII	X	2.4	5.1	2.4	5.1	2.4	5.1	2.4	5.1	2.4	5.1	V
Y5	Converter busy, 3-state (VL)	XXII	X	0	400	0	400	0	400	0	400	0	400	mV
Y6	Converter busy, 3-state (PW)	XXII	X	0.5	6.0	0.5	6.0	0.5	6.0	0.5	6.0	0.5	6.0	µS
Z1-Z3	Power supply ripple rejection	XXIII	Q	← See table XXIII →										

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
AA	Continuous conversion	XXIV	Q	← See table XXIV →										
BB1, BB2	Power supply transient amplitude	XXV	Q	0	700	0	700	0	700	0	700	0	700	mV, pk
CC	Stress	XXVI	Q	← See table XXVI →										

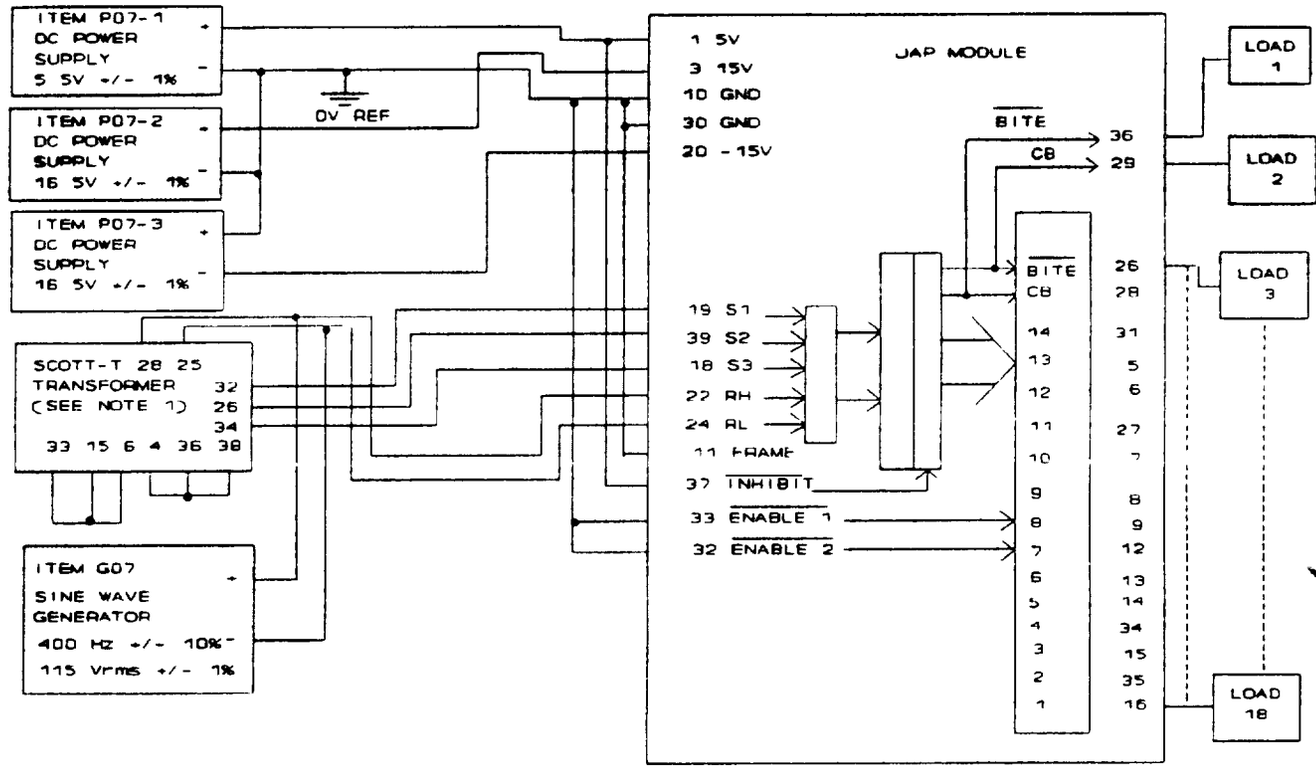
1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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NOTES:

1. Scott-T transformer = M28787/185-1 (Key Code STT).
2. Loads 1 through 18 are as shown on figure 4.

FIGURE 2. Operating temperature, component temperature, and life test circuit.

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TABLE XX. Settling time test.

Test cond	Parameter	Connect			Initial limits 25°C ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
W	Settling time	37	32,33	31	0	200	ms	19, 20

TABLE XXI. Noise sensitivity test.

Test cond	Parameter	Initial limits 25°C ± 5°C ambient		Unit	Test fig
		Min	Max		
X	Noise sensitivity	See note 6 figure 21		--	21

TABLE XXII. Converter busy test.

Test cond	Parameter	Connect	Initial limits 25°C ± 5°C ambient		Unit	Test fig
		A to pin	Min	Max		
Y1	Direct (VH) Converter busy	29	2.4	5.1	V	22, 23
Y2	Direct (VL) Converter busy	29	0	400	mV	22, 23
Y3	Direct (PW) Converter busy	29	0.5	6.0	µS	22, 23
Y4	3-state (VH) Converter busy	28	2.4	5.1	V	22, 23
Y5	3-state (VL) Converter busy	28	0	400	mV	22, 23
Y6	3-state (PW) Converter busy	28	0.5	6.0	µS	22, 23

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TABLE XXIII. Power supply ripple rejection test.

Test cond	Parameter	Connect		Initial limits 25°C ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
Z1	5V Power supply ripple rejection	1	Open	See note 8 figure 24		-	24
Z2	15V "	3	Open			-	24
Z3	-15V "	Open	20			-	24

TABLE XXIV. Continuous conversion test.

Test cond	Parameter	Initial limits 25°C ± 5°C ambient		Unit	Test fig
		Min	Max		
AA	Continuous conversion	See figure 25 note 6		---	25

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